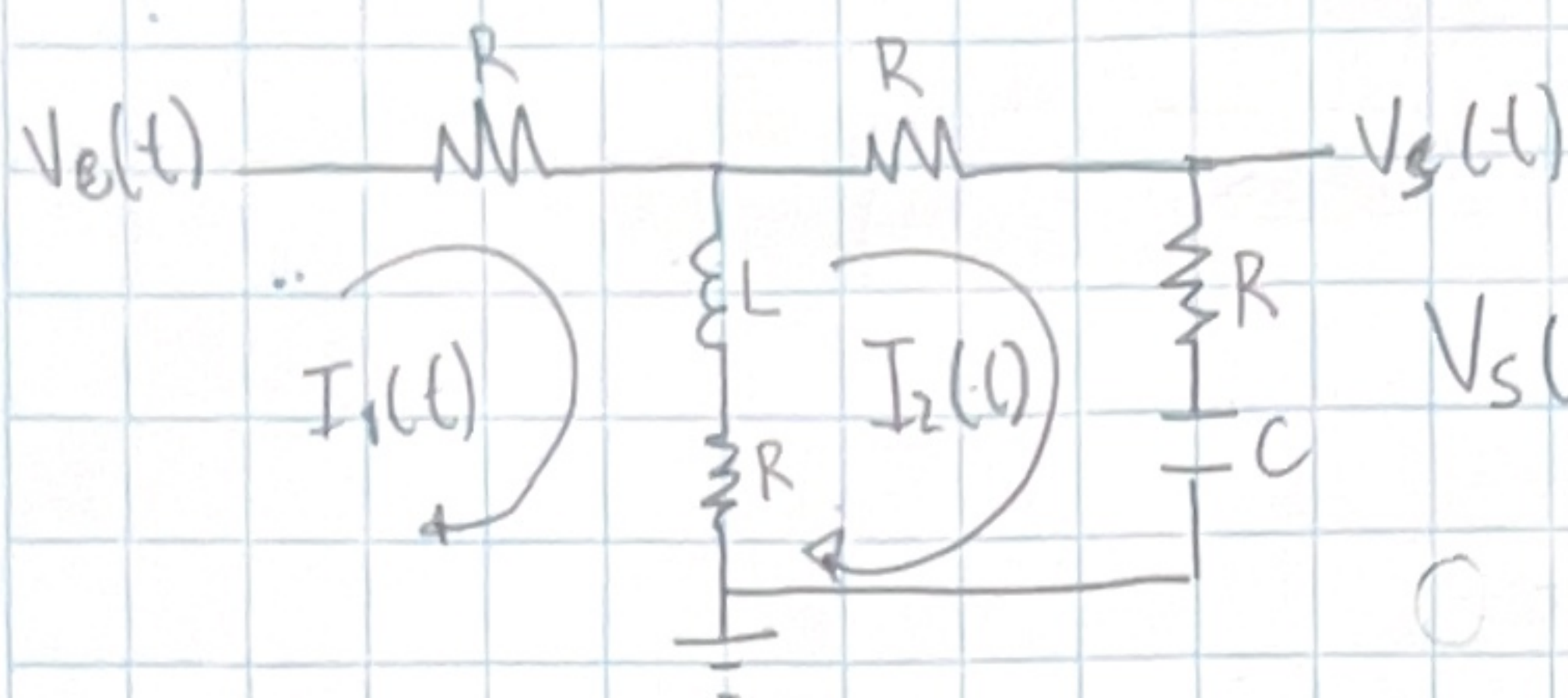


Edoardo Del Toro Nuñez

Modelado en Sistemas Fisiológicos.

C2321239a



$$V_s(t) = RI_1(t) + L \frac{dI_1(t)}{dt} + RI_1(t) - L \frac{dI_2(t)}{dt} - RI_2(t)$$

$$0 = RI_2(t) + L \frac{dI_2(t)}{dt} + RI_2(t) + RI_2(t) + \frac{1}{C} \int I_2(t) dt - RI_1(t) - L \frac{dI_1(t)}{dt}$$

$$V_e(t) = RI_2(t) + \frac{1}{C} \int I_2(t) dt$$

Aplicar Laplace

$$V_s(s) = RI_1(s) + LsI_1(s) + RI_1(s) - LsI_2(s) - RI_2(s)$$

$$0 = RI_2(s) + LsI_2(s) + RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs} - RI_1(s) - LsI_1(s)$$

$$V_e(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = I_1(s) [R + Ls + R] - I_2(s) [Ls + R]$$

$$0 = I_2(s) [R + Ls + R + R + \frac{1}{Cs}] - I_1(s) [R - Ls]$$

$$V_s(s) = I_2 [R + \frac{1}{Cs}] \quad V_s(s) = I_2 \left[\frac{RCs + 1}{Cs} \right]$$

$$I_1(s) = \frac{I_2(s) [3R + Ls + \frac{1}{Cs}]}{R + Ls}$$

$$I_1(s) = I_2(s) \left[\frac{3R + Ls + \frac{1}{Cs}}{R + Ls} \right] \left[\frac{1}{R + Ls} \right]$$

$$I_1(s) = I_2(s) \left[\frac{3R}{R + Ls} + \frac{Ls}{R + Ls} + \frac{1}{RCs + LCs^2} \right]$$

$$I_1(s) = I_2(s) \left[\frac{3R + Ls}{R + Ls} + \frac{1}{RCs + LCs^2} \right]$$

$$I_1(s) = I_2(s) \left[\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} \right] [2R + Ls] - I_2(s) [Ls + R]$$

$$V_e(s) = I_2(s) \left[\left(\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} \right) (2R + Ls) - (Ls + R) \right]$$

$$V_e(s) = I_2(s) \left[\frac{(2CLR^2 + 3CR^2s + 2R + CLs^3 + 3LCRs^2 + Ls) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{2CLR^2 + 3CR^2s + CLs^3 + 2R + Ls - (CLs^3 + CLR^2s + CLR^2s + CR^2s)}{CLs^2 + CRs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{3CLR^2 + 3CR^2s + 2R + Ls + CR^2s}{CLs^2 + CRs} \right]$$

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Modelado de Sistemas Fisiológicos.

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} (I_1(s))}{\frac{3CLR s^2 + 3CR^2 s^2 + 2R + Ls + CR^2}{CLs^2 + CRs} (I_1(s))}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs}}{\frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs}}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

Error Estacionario

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right)$$

$$e(s) = \frac{R}{2R}$$

$$e(t) = 0.5 \text{ V}$$

$$R = 10 \text{ k}\Omega \quad L = 550 \text{ mH} \quad C = 470 \text{ }\mu\text{F}$$

Estabilidad en lazo abierto: Polos del sistema.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

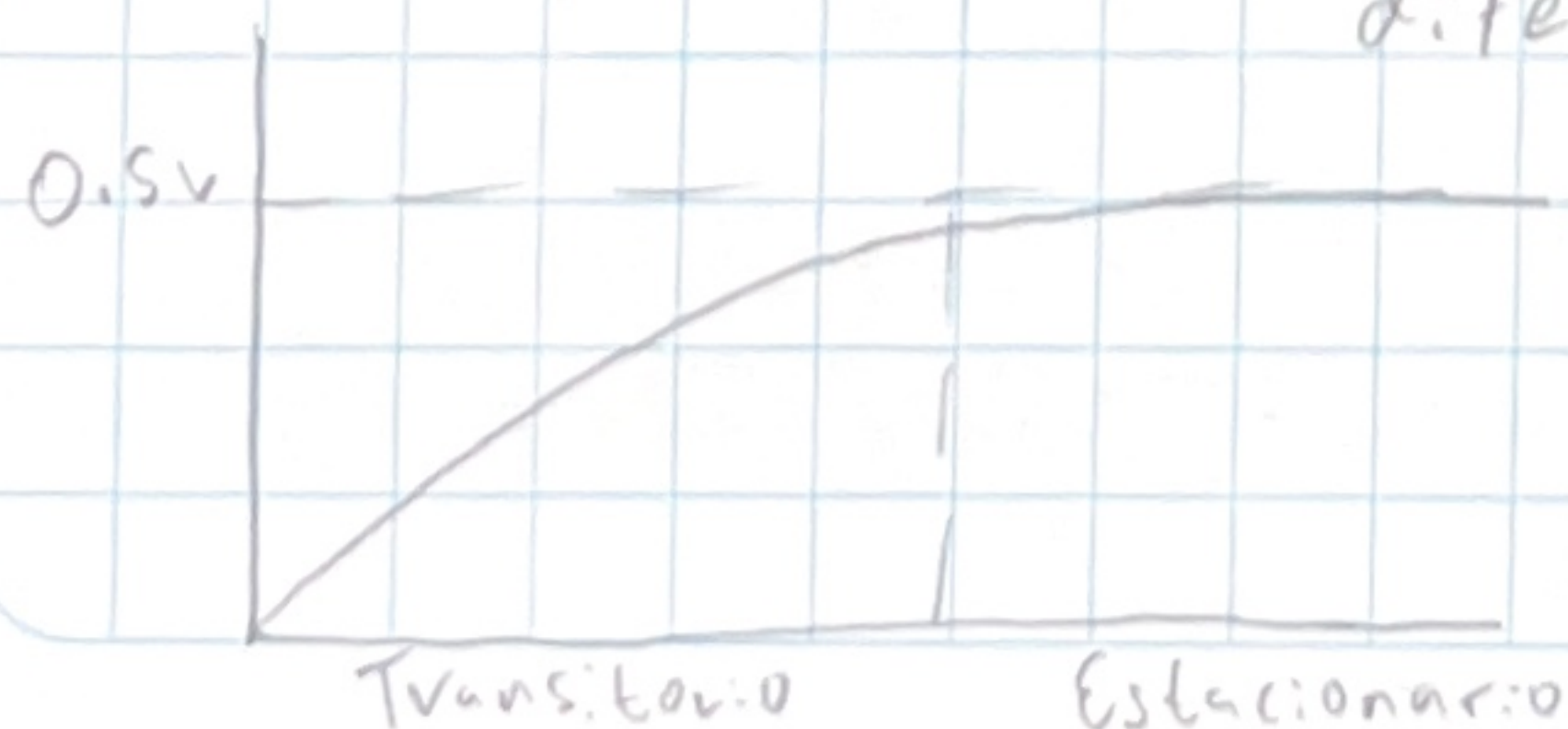
$$a = 7.755 \times 10^{-3} \quad b = 235 \times 10^3 \quad c = 20 \times 10^3$$

$$\lambda_{1,2} = \frac{-235 \times 10^3 \pm \sqrt{(235 \times 10^3)^2 - 4(7.755 \times 10^{-3})(20 \times 10^3)}}{2(7.755 \times 10^{-3})}$$

$$\lambda_1 = -0.085106$$

$$\lambda_2 = -3.0303 \times 10^7$$

Nuestra respuesta es sobreamortiguada porque las raíces son reales negativas con diferente valor.



Modelado de Sistemas Fisiológicos

Modelo de Ecuaciones Integrodiferenciales

A partir de las ecuaciones principales del sistema, se despejan las variables dependientes, es decir, las corrientes y el voltaje de salida.

$$V_s(t) = 2R I_1(t) + L \frac{dI_1(t)}{dt} - L \frac{dI_2(t)}{dt} - R I_2(t)$$

$$2R I_1(t) = R I_2(t) + L \frac{dI_2(t) - dI_1(t)}{dt} + V_s(t)$$

$$I_1(t) = \left[R I_2(t) + L \frac{dI_2(t) - dI_1(t)}{dt} + V_s(t) \right] \frac{1}{2R}$$

$$0 = 3R I_2(t) + L \frac{dI_2(t)}{dt} + \frac{1}{C} \int I_2(t) dt - R I_1(t) - L \frac{dI_1(t)}{dt}$$

$$3R I_2(t) = R I_1(t) + L \frac{dI_1(t) - dI_2(t)}{dt} - \frac{1}{C} \int I_2(t) dt$$

$$I_2(t) = \left[R I_1(t) + L \frac{dI_1(t) - dI_2(t)}{dt} - \frac{1}{C} \int I_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = R I_2(t) + \frac{1}{C} \int I_2(t) dt$$

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Modelado en Sistemas Fisiológicos

$$R = 10 \text{ k}\Omega$$

$$C = 470 \mu\text{F}$$

$$L = 550 \text{ mH}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(470 \times 10^{-6} \cdot 550 \times 10^{-3} \cdot 10 \times 10^3) + (470 \times 10^{-6} \cdot 10 \times 10^3 + 550 \times 10^{-3}) + 10 \times 10^3}{3(470 \times 10^{-6} \cdot 550 \times 10^{-3} \cdot 10 \times 10^3) + s(470 \times 10^{-6}(10 \times 10^3)^2 + 550 \times 10^{-3}) + 2 \cdot 10 \times 10^3}$$

$$\frac{V_s(s)}{V_e(s)} = 0.039234$$